Section 10 bootstrap, with the two open gaps marked

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Assume a smooth divergence-free solution of unforced 3D incompressible Navier–Stokes on $\mathbb{T}^3$ exists on $[0,T)$. Let $M(t) = \|\omega(t)\|_{L^\infty}$ and $M_\varepsilon(t) = \|\omega * \eta_\varepsilon\|_{L^\infty}$.

Auxiliary field.

$$\partial_t \phi + u \cdot \nabla \phi = |\omega|^2, \qquad \phi(0,x) = 0.$$

Along characteristics, $\|\phi(t)\|_{L^\infty} \le \int_0^t M(s)^2\, ds$. This upper bound is linear in t whenever M is bounded below by a positive constant. It is not a uniform constant on $[0,\infty)$.

Pointwise stretching. At an interior maximum of $|\omega_\varepsilon|^2$ the viscous pairing is non-positive, and Calderón–Zygmund gives a bound of the schematic form

$$|\omega_\varepsilon \cdot ((\omega_\varepsilon \cdot \nabla)u)| \le C_{\mathrm{CZ}}(3)\, M_\varepsilon(t)^2,$$

up to the standard logarithmic correction in the Beale–Kato–Majda estimate. The pure quadratic bound without a logarithm is the classical CZ estimate on ∇u in BMO, not a closed L^∞ bound.

Lyapunov inequality, as stated.

$$\frac{dS_\varepsilon}{dt} \le C_{\mathrm{abs}}\big(1 + M_\varepsilon(t)^3 \|\phi_\varepsilon\|_{L^\infty}^{3/2}\big) - \kappa' \int_{\mathbb{T}^3} |\omega_\varepsilon|^6\, d\lambda.$$

Gap A. Substituting the history bound makes the prefactor grow like $t^{3/2}$ if M_ε remains positive. The comparison equation is not the autonomous ODE $y' = Cy^2 - \kappa'' y^3$.

Gap B. The relation $\int |\omega_\varepsilon|^6 \ge c M_\varepsilon^4$ does not follow from Hölder. Hölder supplies $\|\omega_\varepsilon\|_6 \le M_\varepsilon$ on a unit-volume torus. A lower bound requires a lower bound on the measure of $\{|\omega_\varepsilon| \gtrsim M_\varepsilon\}$.

What would close (B). A proof must produce $\int_0^T M(t)\,dt < \infty$ for every finite T, for every smooth divergence-free finite-energy u°, with $f \equiv 0$, and then invoke Beale–Kato–Majda plus parabolic regularity. Neither the tether degeneracy identity $B(F, H) \equiv 0$ nor a holographic analogy supplies that integral bound.